inner current control loop: based on resonant pi control running on master modules

Broadcasted to slaves via distributed control bus

$i_{su}^{ref}(t)$ + − $i_s^u(t)$

$$k_p^{cc} + k_i^{cc} \frac{s}{s^2 + 2\delta\omega_{res}s + \omega_{res}^2}$$

$u_{ctrl}^u(t)$

current control mode: master module cluster phase u

voltage control mode

slave modules

$i_{sv}^{ref}(t)$ + − $i_s^v(t)$

$$k_p^{cc} + k_i^{cc} \frac{s}{s^2 + 2\delta\omega_{res}s + \omega_{res}^2}$$

$u_{ctrl}^v(t)$

current control mode: master module cluster phase v

voltage control mode

slave modules

$i_{sw}^{ref}(t)$ + − $i_s^w(t)$

$$k_p^{cc} + k_i^{cc} \frac{s}{s^2 + 2\delta\omega_{res}s + \omega_{res}^2}$$

$u_{ctrl}^w(t)$

current control mode: master module cluster phase w

voltage control mode

slave modules